Execution-Grounded Hierarchical Graph Learning for Multi-UAV Maritime Data Collection

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Abstract— Multi-UAV maritime data collection couples discrete buoy assignment with continuous motion and finite-queue service. We study a modular hierarchy in which a heterogeneous graph neural network (HGNN) ranks legal assignments using short-horizon execution labels, and a frozen residual multi-agent proximal policy optimization (MAPPO) controller executes each assignment with safety projection. Historical 48-slot benchmarks report mean payload gains of 2.645 Mbit on development validation and 3.502 Mbit on a separate holdout against the same-pool rate baseline. We retain these formal results while extending the analysis to synthetic priority-tagged data. Exact replay of 224 evaluation episodes separates age-weighted utility from actual tagged delivery; subsequent reward, queue-observation, and gradient-clipping interventions do not meet their registered adoption criteria. A frozen-controller assignment diagnosis reveals that candidate-prefix truncation can suppress meaningful alternatives. After matching the participating UAV set, 11 of 96 sampled alternatives improve known tagged delivery without reducing payload beyond a 1-bit tolerance, and five also satisfy the communication and safety screens in two training scenarios. These are local counterfactual opportunities, not gains of a deployable priority-aware policy. The completed fixed small-tag comparison on 16 training states selects the same assignment with and without the 0.001 tag term. That preregistered rule shows no decision effect and is not adopted. A later arrival-label comparison, limited to two already opened training scenarios and four 15-step windows, and a cached idle-reference score change are both closed without adoption. Registered scenarios 322000 and 322001 each match their own saved 48-step Scheme C reference. Two later fixed-budget offline comparisons are not adopted. A layout check repeats scenario 322001 once. These matches are engineering reproduction, not new performance results.

Index terms: multi-UAV systems; maritime data collection; heterogeneous graph learning; residual MAPPO; FIFO queues; priority tags; reproducibility.

I. Introduction

Mobile UAV collectors can connect distributed sensor buoys where fixed infrastructure is sparse. Their assignments determine which queues may be served, while motion changes distances, interference, and the geometry of later choices. A hierarchy separates these decisions: the upper layer chooses a joint assignment, and the lower layer executes it through continuous motion. This separation simplifies interfaces but creates a coordination problem when the upper layer is trained against a short execution window and the final objective is cumulative episode delivery.

Our formal system combines a rate-generated legal candidate pool, an HGNN assignment scorer, and a residual MAPPO executor. The scorer is supervised by the payload produced when each candidate is executed with the frozen lower-level controller. Formal evaluation freezes both modules. The system is therefore a composition of separately trained components rather than an end-to-end jointly optimized agent.

The recent research extension asks a narrower question: can a small preference for explicitly tagged data improve its actual delivery while total payload remains the main objective? Waiting time and priority are different quantities. An increase in age-weighted service does not establish an increase in tagged data delivery. Moreover, if the executor cannot change its assigned buoy, a priority reward may not control which tagged queue is served. We investigate these questions with immutable historical results, exact replay, registered controls, and bounded assignment interventions.

The contributions are: (1) an execution-grounded hierarchical assignment-ranking system with a common frozen executor; (2) scenario-paired historical development and holdout benchmarks; (3) a FIFO bucket-level distinction between age-weighted service and actual priority-tagged delivery, verified by full replay; and (4) an assignment-coverage diagnosis showing how enumeration-prefix sampling can hide local service opportunities. The fourth contribution identifies an experimental-design issue and a bounded opportunity; it does not establish a priority-aware deployment policy.

II. Related Work

A. UAV association and motion

Learning-based UAV trajectory design has considered data collection, collision avoidance, connectivity, and interference [1], [2]. Queue-aware maritime graph reinforcement learning additionally studies buffered collection and feasible mixed association decisions in a richer sensing-and-communication setting [5]. Our simulator and objective are narrower: two-dimensional movement, fixed sensor buoys, simplified propagation, and exact delivered payload. We do not compare numerical gains across these different physical models or objectives.

B. Graph representations and modular control

MAPPO provides a centralized-training framework for cooperative agents [3]. Heterogeneous graph learning represents typed nodes and relations [6], while listwise learning emphasizes relative candidate order [7]. We use these ideas for assignment scoring with a fixed executor; the architecture is not claimed to reproduce the full Heterogeneous Graph Transformer. Waypoint Transformer studies intermediate targets in supervised reinforcement learning [4]. The repository derives from that codebase, but the current formal maritime decision path is the HGNN-MAPPO hierarchy described here, not a Transformer motion policy.

C. Evidence from diagnostic interventions

A counterfactual candidate rollout answers what could happen from a particular state under a chosen assignment. It does not answer whether a causal online selector would choose that assignment, or whether repeated use would improve a complete episode. We keep these levels of evidence separate throughout the priority-aware extension.

III. System Model and Formal Hierarchy

A. Environment and exact payload

The canonical configuration has six UAVs, twelve buoys, a 160 m by 80 m region, a 48-slot horizon, a 15-slot task hold, a maximum displacement of 22 m per slot, and a nominal service radius of 16 m. Bandwidth is 1 MHz, the path-loss exponent is 2.2, and same-cluster/cross-cluster interference factors are 1.0/0.08. These are simulator parameters, not measurements from a maritime deployment. The code obtains entity counts and dimensions from configuration.

$$p_m(t+1) = \Pi_\Omega(p_m(t) + Lu_m(t)),$$
$$\|u_m(t)\|_2 \le 1.$$

A joint assignment is a legal partial injection from UAVs to buoys; -1 denotes an idle UAV, and no non-idle buoy is assigned twice. Actual service requires an eligible link and nonempty queue. The channel accounts for concurrent links and cluster-dependent interference. Service occurs before new arrivals are admitted.

$$d_k(t) = \min(q_k(t), c_k(t)),$$
$$q_k(t+1) = q_k(t) - d_k(t) + a_k^{\mathrm{adm}}(t).$$

$$P = \sum_{t=0}^{47} \sum_k d_k(t).$$

Here accepted arrivals already respect finite capacity. Generated, dropped, or queued bits are not delivered payload. Numerical queue snapping is tracked separately and must not be credited as service. The formal primary objective is expected episode payload under association, movement, queue, and safety constraints.

B. Legal candidates and HGNN ranking

At decision times 0, 15, 30, and 45, the formal generator retains each UAV's top six buoy targets by current hypothetical rate and enumerates legal joint assignments, including idle options. The parameter k is a per-UAV target limit, not the number of joint assignments. The reported canonical pool has 13,327 assignments; this count should not be assumed universal across other geometries or configurations. Stable ordering fixes ties and score-to-assignment binding.

The graph contains UAV and buoy nodes, UAV-buoy and UAV-UAV edges, and global context. Formal tensors include UAV [1,6,8], buoy [1,12,8], pair edges [1,6,12,4], UAV-UAV edges [1,6,6,2], and global [1,3]. The encoder exchanges typed messages and conditions on global context. Future payload labels, future SINR, utility ranks, and subsequent delivered bits are excluded from inference inputs. A matching decoder sums selected pair scores; idle terms contribute zero.

$$S(a_i) = \sum_m \mathbf{1}[a_{i,m} \ge 0]\, f_{\mathrm{pair}}(h_m^U, h_{a_{i,m}}^B).$$

The selected formal scorer is Scheme C Epoch 1, initialized from the HGNN Epoch 6 checkpoint. Its matching decoder remains frozen while the encoder is updated. Candidate labels are the 15-slot executed payload of frozen MAPPO-V2 from a restored state. The original high-level labeling protocol disables future arrivals, unlike the later priority diagnoses, which preserve the natural arrival/tag stream. This protocol difference is retained explicitly.

$$q_i = \frac{y_i - \min_j y_j}{\max(\max_j y_j - \min_j y_j,\, 10^{-6})},$$
$$p = \mathrm{softmax}(30(q - \max q)),$$
$$\mathscr{L} = -\sum_i p_i \log \mathrm{softmax}(S)_i.$$

The label vector is detached before this target is formed, and the resulting distribution p is detached from the computation graph. The denominator is the larger of the label range and 10^{-6}. Subtracting the maximum quality before the temperature-30 softmax is the implemented target; it matches a normalized exponential after that shift.

C. Residual motion and frozen interfaces

TaskOption binds the selected assignment for up to 15 slots. The lower-level actor observes self, task, communication, and neighboring-UAV features and produces a two-dimensional residual around a fly-toward prior. Deterministic evaluation uses the actor mean, with residual scale 0.25.

$$\delta_m(t) = 0.25 \tanh(\mu_m(t)),$$
$$u_m^{des}(t) = \mathrm{clip}_{unit}(u_m^{fly}(t) + \delta_m(t)).$$

SafetyProjector processes the joint motion command. A constraint failure or non-finite projection triggers the registered zero-action fallback and is counted. Projection interventions, constraint failures, and measured safety violations are distinct outputs. The low-level actor cannot rematch targets within a hold.

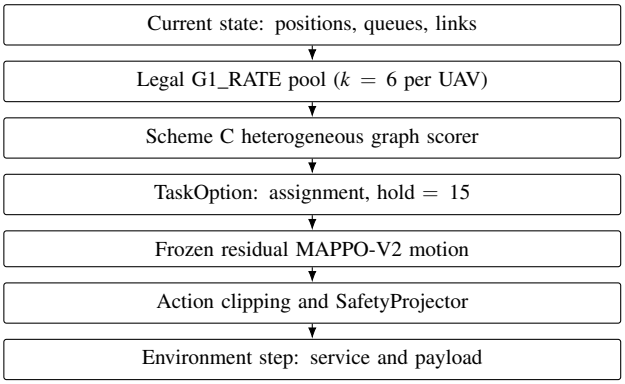


Figure 1: Formal path: current state, G1_RATE k=6 pool, Scheme C HGNN, TaskOption, frozen MAPPO-V2, SafetyProjector, and exact queue service. The priority diagnoses use a separate historical k=5 sampler and do not replace this path.

$$r_{\text{base}}(t) = D(t)/10^6 - 0.05c_{\text{project}}(t) - 0.01c_{\text{residual}}(t).$$

D(t) is actual delivered bits. This shaped training reward is not the episode payload metric. Formal inference continues to use phase6g_seed0.pt; none of the subsequent priority-aware variants replaces it.

IV. Priority-Aware Accounting and Experimental Design

A. Synthetic tags and FIFO provenance

A separate ledger attaches a binary synthetic tag z to each admitted bucket. Tags do not represent verified maritime business urgency. The environment still serves FIFO: buckets are ordered by admission step and insertion index, and ordinary data ahead of tagged data cannot be skipped. Initial reset queues are ordinary. Only actually served bucket bits contribute to tagged delivery; drops, arrivals, snapping, and residual backlog do not.

$$u_b(t) = \min(1, \min(\text{age}_b(t)/48, 1) + 0.25z_b).$$

Let A be delivered bits weighted by the clipped age term, I the incremental weight added by the tag after clipping, and U their combined weighted service. Then U=A+I by construction. Actual tagged delivery B is instead the sum of z times delivered bucket bits. A, I, and U are bit-equivalent weighted quantities; they must not be presented as actual delivered tagged bits.

The early bounded bonus limits additional reward relative to payload. A later registered experiment separates the age and direct-tag terms: min(0.0005 A/10^6, 0.00025 D/10^6) plus min(0.0010 B/10^6, 0.00075 D/10^6). These coefficients describe an unsuccessful research variant, not a new formal objective or an optimized choice.

Table I: Evaluation scopes; rows are not interchangeable.

Study	Scenarios / unit	Interpretation
Formal hierarchy	322048-322063 / episode	Development validation; used for selection
Formal hierarchy	322064-322079 / episode	Historical separate holdout
Priority training	321000-321095	Training pool; not an independent holdout
Priority evaluation	321096-321127 / episode	Repeatedly used development pool
E17 / E18	321000-321007 / state	8 training scenarios; t=0,15,30
E19 completed	321008-321023 / state	T and M select the same assignment; no decision effect

B. Separate scenario namespaces and evidence levels

The formal high-level training split is 322000-322047; its dataset samples t=0,15,30. The priority reward/observation/clipping experiments use three model seeds and the 32-scenario low-level development pool. Repeated use of that pool precludes treating later comparisons as fresh generalization evidence. E17/E18 diagnose the historical G1_RATE k=5 sampler with frozen L0, not the formal Scheme C k=6 planner.

C. Reproducibility and pairing

Priority replay explicitly reapplies one PyTorch thread, deterministic algorithms, and process seed 2 before each evaluation episode. Seed 2 reconstructs a historical inherited process context; it is not a universal model or evaluation seed. Environment reset uses the scenario ID, while assignment RNG uses 7000000 plus that ID. Model seed identifies the trained checkpoint. The formal AB/AC benchmark uses its own arm seed convention, 1000000 plus scenario, and is not relabeled with the priority replay context.

Each assignment branch restores environment and RNG state, ledger, TaskOption, hold counter, previous motion, assignment RNG, and process RNG. Common generated arrivals and tags are verified. A branch first performs the baseline draw to align the assignment stream, then installs the registered association. Payload tolerance is 1 bit, path tolerance 10^{-9} m, and bucket-attribution tolerance 10^{-6} bit. Tolerances are not enlarged to turn a failed replay into a pass.

Episode confidence intervals use scenario-paired differences, with 10,000 bootstrap draws and seed 0 where specified by those historical protocols. E18 candidates share states and scenarios: their counts are diagnostic screens, not independent replicates. No candidate-level confidence interval or new performance PASS is inferred from those counts.

V. Results

A. Historical formal hierarchy benchmark

Both arms use the same frozen MAPPO-V2 executor and G1_RATE k=6 legal pool. Historical records support a positive average difference in these measured scenarios, not universal

Table II: Historical 48-slot results retained from the original paper and system overview; Mbit.

Metric	Validation	Holdout
Number of scenarios	16	16
Scheme C mean payload	68.918	69.151
Rate baseline mean payload	66.273	65.648
Mean paired difference	+2.645	+3.502
Paired 95% interval	[0.180,5.017]	[1.270,5.395]
Positive / tie / negative	11 / 0 / 5	14 / 0 / 2

dominance. The two negative holdout cases are 322065 (-8.162 Mbit) and 322071 (-1.871 Mbit). The holdout was separate from checkpoint selection; the current priority studies do not reuse it to tune tags.

The original manuscript's offline table reports selected J15 increasing from 18.892 to 19.240 Mbit, Top-1 from 0.5625 to 0.6667, Top-3 from 0.6250 to 0.7083, Top-5 from 0.6667 to 0.7083, and normalized regret from 0.0507 to 0.0355. The dataset protocol has 48 validation decision states from 16 scenarios at three times, correcting the original prose's description of a 16-state set. The selected side is checked against the original saved aggregate at best epoch 1 in the archived Scheme C result: 19240091.24202323 bits (19.240091242 Mbit), Top-1 0.6666666666666666, Top-3 and Top-5 0.7083333333333334, and normalized regret 0.035473433715573995. Rounded to the digits already printed above, these are 19.240, 0.6667, 0.7083, and 0.0355. The label is a check of that saved aggregate, not a recomputed statistic. The baseline side, beginning at 18.892 Mbit, remains reported-only because those cells are not in the same saved aggregate. The missing top-level summary and the baseline and best per-state series stay missing and are not back-calculated. The original result stores a non-numeric loss; a separate validation-loss backfill is a different field and is not substituted for it.

The archived AB and AC summaries from commit `8a34efc718665a7cfe4375b662e8f5249263f03d`, together with their 32 episode files, reproduce Table II after rounding payload to three decimal Mbit. The development summary is marked validation_is_development_set. The holdout file is a saved result and was not rerun; its two negative cases remain 322065 (-8.162 Mbit) and 322071 (-1.871 Mbit). Episode payload fields match the corresponding summary rows. The ranking-loss summary.json is still absent. Only the Scheme C side of the offline ranking sentence is traced to the saved best-epoch aggregate; the baseline side stays reported-only. The unrelated FINAL-HIER paper_assets tables are not substituted for these benchmarks.

B. Exact replay changes the urgency interpretation

E9 replays all 224 saved evaluation episodes under the reconstructed context, with maximum reported payload and path differences of zero and exact age-plus-tag reconciliation. For L0, aggregate A, I, and U are 436.047, 86.900, and 522.947 million bit-equivalents, respectively, while actual tagged delivery is 350.303 Mbit. These aggregates have different units of interpretation even though each is expressed on a bit scale.

Table III: Priority research outcomes; no listed variant is adopted.

Stage	Registered comparison	Outcome
E10	Separate age/direct-tag reward; 3 seeds	0/3 pass; FAIL
E12	Actor queue lookahead vs masked inputs	0/3 pass; FAIL
E14	Union vs split clipping pilot	Measurable Adam-update difference only
E15	Full union/split control; 3 seeds each	0/3 pass; INCONCLUSIVE
E16	Clipping-line reconciliation	Closed without adoption

Across L2 seeds, age-weighted paired gains are about 0.845-0.940 million bit-equivalents per episode. Direct tagged-delivery means are instead -0.239, -0.346, and -0.154 Mbit, each with a 95% interval crossing zero. Thus the apparent urgency improvement is primarily attributable to age weighting in this experiment, without evidence of a reliable actual-tagged gain. Near-complete retention of the tag increment also argues against clipping saturation as the main explanation in these saved trajectories.

C. Registered low-level interventions

The direct-tag reward experiment has negative tagged-delivery means for all three seeds, and two seeds increase registered safety counts. Queue-lookahead inputs expose tagged backlog behind ordinary FIFO heads but do not meet the benchmark criteria. These results reject the tested configurations within their development protocol; they do not prove that priority-sensitive control is impossible.

The clipping study compares a joint actor/critic norm bound G=0.5 against separate bounds G/sqrt(2), preserving the combined bound. The pilot measures a change in actual Adam updates, which is more informative than inferring optimizer behavior solely from a clipping coefficient. The full comparison uses 1,800 PPO updates, 28,800 Adam steps, and 224 evaluation episodes. Tagged-service paired means are positive in the split group, but their intervals include zero against both the same-seed joint control and L0; two payload means against the joint control are negative. The scientific gate remains inconclusive, and the line is closed without adoption.

Training and evaluation safety records must remain separate. E15 records 40 training safety events and one constraint failure; zero safety/constraint events in E18's frozen candidate replay do not erase those events. Prior audits also show different numerical tolerances at the safety counter and projector boundary. Original counts are retained rather than waived retrospectively.

D. Assignment coverage: E17 versus E18

E17 completes 2,544 environment steps over 24 snapshots and 120 candidate rows, with no registered S0/S1/S2 opportunity. Its rule retains the baseline and the first four unique alternatives in enumeration order. Inspection shows that 24 nonbaseline alternatives idle every UAV and 72 activate only one, while the

Table IV: E18 nested screens over 96 sampled alternatives.

Screen	Alternatives	Distinct scenarios
S0: tag gain + payload floor	11	5
S1: plus safety/constraints	11	5
S2: plus SINR/interference	5	2

natural baselines activate two to five UAVs. The executor followed the registered prefix rule; the coverage defect is in the sampling design. A zero within-prefix result does not establish infeasibility of the full joint pool.

E18 retains the same eight training scenarios, snapshots, frozen executor, k=5 generator, hold, and event accounting, but restricts alternatives to exactly the baseline's active UAV mask. This mask identifies which UAVs have nonnegative target assignments; it does not assert that those UAVs are in radio range or successfully serving. Alternatives are ordered by original pool index, split into four contiguous strata, and sampled with an independent fixed per-snapshot RNG. The selection manifest is committed before alternative replay.

E18 finds 2,696 eligible same-mask unique alternatives summed over the 24 states and evaluates a registered sample of 96, plus 24 baselines. All states have four interpretable alternatives. The total budget is 384 reference steps, 1,800 candidate steps, and 360 repeated-baseline steps, again 2,544. Reference reconciliation and t=0 negative controls pass.

Service attribution distinguishes buckets already admitted at the snapshot from future-admitted buckets. The main gain is in snapshot-known tagged service. S0 requires a gain greater than 1 bit with payload difference at least -1 bit. S1 additionally requires nonincreasing raw safety and constraint counts. S2 additionally requires nondecreasing mean SINR and nonincreasing mean interference.

All S0 gains occur at t=15. Known tagged service increases by approximately 1-2 Mbit; payload differences among S0 rows range from 0 to 9 Mbit. The S2 rows occur in 321002 and 321006, with payload increases of approximately 4.160-9 Mbit. All 120 candidate rows have zero raw safety and constraint events. Multiple successes in one scenario are not independent replications.

The registered result is replicated local opportunity, not a learned-policy gain. Matching the active mask omits opportunities that require changing participating UAVs; sampling omits most legal assignments; the comparison omits t=45 and other scenario families. Selecting a best candidate after replay remains a hindsight diagnostic. In particular, these k=5 results neither replace nor re-evaluate the formal Scheme C k=6 benchmark.

VI. Completed Fixed Small-Tag Comparison

E19 is now a completed diagnostic at result commit `f9e9a9e007d7d84046232181fdab6f9a732a4a05`. This manuscript cites that fixed report and does not rerun the comparison. The rule was not adopted, and it did not train a new model.

E19 fixes 16 training scenarios, 321008-321023, and a single t=15 snapshot per scenario. They are additional to E17/E18 but may have appeared in earlier low-level training, so they are not an independent unseen holdout. From each natural baseline draw N, the selector considers only the original same-mask k=5 pool, including N.

$$\widehat{c}_{mk} = \min\{q_k, [15 - \eta_{mk}]_+ \Delta t\, \widehat{r}_{mk}\},$$
$$\widehat{P}(a) = \sum_m \widehat{c}_{m,a_m}.$$

The rate and geometric ETA are existing current-state features. A FIFO scan of current admitted buckets determines the tagged bits K-hat within the estimated service prefix. This is a static service proxy, not a joint-interference forecast or true-payload guarantee. The tag-masked rule M maximizes P-hat; rule T maximizes P-hat + 0.001 K-hat, with identical candidate and tie-breaking logic. Because 0 <= K-hat <= P-hat, the added score is at most 0.1% of the base proxy. This coefficient is fixed before evaluation and does not change the formal reward.

The selector receives no future arrival/tag information, scenario-ID feature, candidate outcome, or hindsight best index. Masking affects only M's ranking input; actual accounting keeps the true synthetic tags. All selected assignments are committed before alternative replay. T-M is the primary tag ablation; T-N checks the natural baseline; M-N describes the effect of base sorting.

The registered ceiling was 1,728 environment steps. The completed comparison reports 1,488 formal replay steps, zero training updates, and zero parameter updates. Thirty voided steps from the earlier replay are also part of the accounted consumption, so the accounted total is 1,518. The figure 1,488 counts the formal replay. On all 16 states, T and M select the same assignment, so the number of differing scenarios is 0. The mechanism evidence is NO_TAG_DECISION_EFFECT. The T-M known-tag service difference and the T-M payload difference both have mean 0, median 0, and a 95% interval of [0, 0]. performance_pass, adopted, release_eligible, and merged are false. The scientific gate is FIXED_TAG_RULE_DIAG_COMPLETE. T-N and M-N payload intervals include zero and are not a performance pass. An identical T/M choice is a zero decision effect for this fixed 0.001 coefficient; it is not a reason to increase the weight, enlarge k, or move to development or test scenarios. This single-window sample does not establish full-episode policy improvement.

VII. Closed Arrival-Label Checks

After E19, a separate maintenance chain asked whether real-arrival labels or an idle-reference score should replace the formal selector. It is closed without adoption. The checks use two already opened training-pool scenarios, 322022 and 322023, and four 15-step windows at $t = 15$ and $t = 30$. They do not open a new holdout, and they do not measure a 48-slot policy gain.

L3, commit `8052e96267aa0699e30b6b37927f4971c8ba51b4`, trains two encoders for 240 supervised updates on 12 states. On the four windows, the real-arrival arm averages

Table V: Unadopted fixed-budget offline payload differences, in Mbit. Exact bit values are in evidence_20261109.json.

Comparison	Overall	t=15	t=30
L8 R−L3B10	+4.006856	+8.044598	−0.030885
L8 R−E6	+0.553303	—	—
L9 R28−R12	+0.789951	+1.090576	+0.489326
L9 R28−L3B10	+4.353406	—	—
L9 R28−E6	−0.157343	—	—

20.150 Mbit, the zero-arrival arm averages 25.905 Mbit, and the Epoch-6 initialization averages 25.863 Mbit. The comparison stays inside that candidate subset.

L4, commit `d25e97fec756028a394a1cca67edb6d29c47c137`, scores the frozen initialization and the two trained checkpoints once, with 48 forwards and no further training. Its edge summary uses the later correction: 36 unique legal edges that appear among the 128 candidates. A within-group regret of zero is not ranking evidence when the chosen group contains only one candidate.

L5, commit `30c880bedd9da4adff0b2de1d6b3fbef4bc27554`, changes no weights. Replacing the additive score by a row-mean idle reference raises the real-arrival arm's four-window descriptive mean from 20.150 to 24.404 Mbit. The same rule has negative differences at $t = 30$ and lowers activity and payload for the initialization and zero-arrival controls at $t = 15$. Neither selector is adopted. This is not a generalization pass.

L6, commit `77c73c00462ad2bb3d778d12a2ad98f0e4ffd576`, adds a single-scenario formal command. Its 48-step Scheme C run on scenario 322000 matches that scenario's saved reference, including payload and all 48 served increments within 1 bit.

L8, commit `77eb078e54dde9e6ca0a55c3e47a3c482ff20d2c`, and L9, commit `bbb7a0eb329fc83da7cd59d91310da198def3b7b`, are later fixed-budget offline comparisons. Table V reports their payload differences in Mbit. The ListNet arm is L3B10, not Scheme C Epoch 1. Both comparisons use historical training states, 128 candidates, and 15-step scores on MAX_RATE_INDEPENDENT snapshots. They are not an independent generalization test and not a 48-step formal gain. L8 is NOT_PROMOTED because its t=30 difference fails the preregistered -1 bit condition, even though the overall difference versus Epoch 6 is positive. L9 is NOT_PROMOTED because R28 is below Epoch 6. L3–L5 stay closed. Neither comparison is adopted, and the formal model is unchanged.

L10, commit `8fc6fda3d50d51b414cb609b7a8953951788eb18`, binds scenarios 322000 and 322001 to their own saved references. Each matches its own reference, and the cross comparison remains different. L10 adds one consistent 48-step run of 322001. The 9 November layout check repeats that scenario once through the command `python-mwt.cli.evaluate`. The repeated run has payload 68952073.58225587 bits, served bits 68952073.58225589, process seed 1322001, reset seed 322001, decisions at 0/15/30/45, four high-level calls, and 48 low-level steps. It matches within 1 bit. These runs are engineering reproduction. They do not change the official policy.

VIII. Discussion and Limitations

A. What the combined evidence supports

The historical formal benchmark supports a positive average payload difference for Scheme C in its measured configuration. The archived AB and AC summaries now match Table II. The Scheme C side of the offline ranking sentence matches the saved best-epoch aggregate, while the ranking-loss summary and the baseline side of that sentence remain unresolved. The priority evidence is narrower: replay separates age effects from actual tagged delivery; the tested low-level variants fail their adoption criteria; state-matched assignment alternatives reveal local room for improvement; and the completed E19 selector makes the same choice with or without the fixed tag term. The later arrival-label and idle-reference checks are also not adopted. The L8 and L9 fixed-budget comparisons are not adopted. No new high-level or low-level policy is adopted.

B. Remaining limits

The environment is synthetic, planar, and small, with fixed buoy geometry and simplified interference. Neither the formal holdout nor the priority tags are field validation. The hierarchy is separately trained and candidate-limited. Zero-arrival 15-slot labels can disagree with natural-arrival 48-slot objectives. FIFO blocking limits the immediately accessible tagged workload, while residual motion cannot change its target during a hold.

The priority development pool has been used repeatedly. E18 has only eight training scenarios and finite same-mask samples, with opportunities concentrated at t=15. It provides no global upper bound and no estimate of a deployable selector's expected episode gain. A nominally safe projector does not remove the need to inspect raw event counts and numerical boundary effects.

Reproducibility remains incomplete at the repository level. The E18 clean-baseline comparison records 206 historical failure signatures in both baseline and changed code, with no new failures and passing related tests. This is not a fully passing repository suite or release qualification. The AB and AC summaries and their episode files are recovered and checked. The Scheme C best-epoch aggregate is now traced, while summary.json and the per-state ranking series are still missing. The recorded runtime is not a newly verified installation, and the historical requirements.txt pins remain different from that recorded runtime.

C. Research direction

The fixed E19 selector does not produce a decision difference against its masked control on the 16 registered states. That result does not justify a larger tag weight. The later arrival-label and idle-reference checks are closed without adoption and do not replace the formal planner. Registered scenarios 322000 and 322001 each match their own saved reference. The additional 48-step runs are engineering reproduction and do not establish a new gain. New low-level training, clipping searches, and formal-planner replacement are not implied by these local results.

IX. Conclusion

We have presented a modular execution-grounded hierarchy for multi-UAV maritime collection and revised its evidence base through the completed E18 diagnosis. Historical formal records report positive average 48-slot payload gains of Scheme C with frozen MAPPO-V2. The priority extension establishes that age-weighted utility is not actual tagged delivery and that several registered low-level changes do not meet adoption criteria. Correcting a candidate-prefix coverage defect reveals local, safety- and communication-screened assignment opportunities under a frozen executor. The completed 0.001 tag rule does not change any registered choice relative to its mask. The later arrival-label and idle-reference checks are not adopted. L8 and L9 are not adopted. Registered scenarios 322000 and 322001 match their own saved references, and the repeated 322001 run adds no new performance claim. The formal system is unchanged, and E19 is not adopted.

Reproducibility and Artifact Availability

The E18 evidence snapshot remains repository revision `26e35e9abcf2d79a7e5d3b932387af8d32d88431`. The completed E18 result commit is `c6aac5b71f23a33653e21db1f2d558f212167be3`; implementation and pre-replay selection commits are `f79ad10139eacbb298b4a306a2b917057b32ef6b` and `83fe1476f0dd109e49425db67dfa50ead5638445`. The completed E19 result commit is `f9e9a9e007d7d84046232181fdab6f9a732a4a05`. L8 is `77eb078e54dde9e6ca0a55c3e47a3c482ff20d2c`, L9 is `bbb7a0eb329fc83da7cd59d91310da198def3b7b`, and L10 is `8fc6fda3d50d51b414cb609b7a8953951788eb18`. This 9 November revision records one engineering consistency run of scenario 322001 and no training. It does not open a holdout. The formal weights are `checkpoints/formal/scheme_c_epoch1.pt`, `checkpoints/formal/mappo_v2_seed0.pt`, and `checkpoints/formal/matching_decoder.pt`. The active policy is `configs/formal_scheme_c_policy.json`.

The formal low-level SHA-256 is `92aa68785c898415439ad8c6480427593e10c851e3d392ee3a8110f3a58dbd0e`; Scheme C Epoch 1 is `f77c0bcef3f359e01de35b75753f3776a9e8332bf65fd191f112236df074f864`. E18 records Python 3.14.2, NumPy 2.4.1, PyTorch 2.12.0+cpu, one thread, and deterministic algorithms. This runtime record is not a verified replacement for the repository's older requirements.txt.

The 2 November index remains evidence_20261102.json. The 4 November check of the recovered AB/AC summaries and episodes is evidence_20261104.json. The 8 November index traces the Scheme C best-epoch aggregate in evidence_20261108.json and leaves the missing summary and per-state series unresolved. evidence_20261109.json records the L8 and L9 bit values, their source commits, and the post-migration paths. The 2 November, 4 November, and 8 November manuscripts and PDFs are retained unchanged. Build status for this 9 November source is recorded in the source manifest from the actual compiler result.

References

[1] S. Xu, X. Zhang, C. Li, D. Wang, and L. Yang, "Deep reinforcement learning approach for joint trajectory design in multi-UAV IoT networks," IEEE Trans. Veh. Technol., vol. 71, no. 3, pp. 3389-3394, 2022. [source].

[2] X. Wang and M. C. Gursoy, "Learning-based UAV trajectory optimization with collision avoidance and connectivity constraints," IEEE Trans. Wireless Commun., vol. 21, no. 6, pp. 4350-4363, 2022. [source].

[3] C. Yu et al., "The surprising effectiveness of PPO in cooperative, multi-agent games," Advances in Neural Information Processing Systems, vol. 35, 2022. [source].

[4] A. Badrinath, Y. Flet-Berliac, A. Nie, and E. Brunskill, "Waypoint Transformer: Reinforcement learning via supervised learning with intermediate targets," Advances in Neural Information Processing Systems, vol. 36, 2023. [source].

[5] B. Li et al., "Queue-aware graph reinforcement learning for UAV-ISAC-assisted maritime data collection," arXiv:2607.00324v2, 2026. [source].

[6] Z. Hu, Y. Dong, K. Wang, and Y. Sun, "Heterogeneous graph transformer," Proc. The Web Conference, pp. 2704-2710, 2020. [source].

[7] Z. Cao, T. Qin, T.-Y. Liu, M.-F. Tsai, and H. Li, "Learning to rank: From pairwise approach to listwise approach," Proc. ICML, pp. 129-136, 2007. [source].

[R0] Original six-page manuscript, retained unchanged. [source].

[R1] Historical formal system overview. [source].

[R2] E9 exact replay and urgency attribution report. [source].

[R3] E10 direct-tag reward report. [source].

[R4] E12 queue-lookahead report. [source].

[R5] E15 full clipping comparison and E16 closure. [source].

[R6] E17 feasibility report. [source].

[R7] E18 matched-mask diagnosis and linked raw artifacts. [source].

[R8] E19 completed fixed-tag diagnosis; T and M selections are identical. [source].